# Lukas Raad


Personal Information:

Teglværksgade 27B
2100, København Ø
Nationality: Danish
Date of birth: 08/01/2002

Contact information:

Mobile: +45 50 26 32 23
E-mail: lukasraad@gmail.com
s204701@dtu.dk


## EDUCATION

**Technical University of Denmark** — **Kongens Lyngby, Denmark**
*MSc Mathematical Modeling and Computation, Focus area ML and Data Science* — *February 2024 – July 2026*
*BSc General Engineering, Specialization in Future Energy* — *September 2020 – December 2023*

**Cornell University** — **Ithaca, New York**
*Semester Abroad, Electrical and Computer Engineering* — *August 2022 – December 2022*

**Lycée Français MLF de Bahreïn** — **Muharraq, Bahrain**
*French Baccalaureate, Scientific Option* — *September 2018 – June 2020*

## WORK EXPERIENCE

**Blue Power Partners** — **Copenhagen, Denmark**
*Junior Analyst* — *June 2024 – July 2026*

- Worked as a junior analyst in the Modeling department of a renewable energy consultancy.
- Developed and implemented new features to the existing optimization code base used in the department.
- Developed a price forecasting model to predict power prices that were used in key projects in various departments.
- Experimented with ML Approximations for LP models.

**Cobham Satcom** — **Kongens Lyngby, Denmark**
*Student Assistant* — *May – July 2022*

- Worked as a student assistant in the shipping department of a leading manufacturer of satellite terminals.
- Assisted in the implementation of a new task management solution for IT issues within the department.
- Produced weekly reports, updated databases, started slide decks.

## PROJECTS & CLUBS

**Federated Learning** — **February – June 2025**

- Special course researching federated learning as a framework for machine learning.
- Implemented and tested different federated learning algorithms such as FedAvg, FedPer and SCAFFOLD.

**Reproducing Visual Decoding and Reconstruction via EEG Embeddings with Guided Diffusion** — **February – June 2025**

- Reproduced the results of "Visual Decoding and Reconstruction via EEG Embeddings with Guided Diffusion".
- Tested the impact on the performance of the model by pretraining on all subjects and finetuning on a smaller set per subject.

**Bachelor Project** — **September - December 2023**

- Designed a mathematical model for the signals emanating from cantilevered piezoelectric energy harvesters.
- Gained a deep understanding on the requirements for vibration based energy harvesting systems using piezoelectrics.

**Vermillion Racing Team** — **March – May 2022**

- Participated in a team effort to develop a car to enter in a competition.
- Developed a brake light design and assisted the communication between the microcontrollers in the electrical system.

## SKILLS

- Python for ML and Data Analysis
- MATLAB for data visualization and analysis
- Image Analysis
- Time series analysis
- Git
- MLOps

## LANGUAGES

**English** (Fluent) | **French** (Fluent) | **Arabic** (Fluent) | **Spanish** (A2) | **Danish** (B1)